Priya Kumari

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Education

Indian Institute of Technology (IIT) Mandi Sep 2024 – May 2026

- **M.Tech** – Computer Science & Engineering (Pursuing)

YMCA University of Science & Technology, Faridabad Sep 2018 – May2022

- **B.Tech** – Computer Science & Engineering | **CGPA: 7.73**

Skills

- **Languages: Python , Java, SQL**
- **AI/ML & Deep Learning**: Deep Learning, Transformer Models**, Large Language Models , Vision-Language Models** (VLMs), Multimodal AI, Retrieval-Augmented Generation , Model Evaluation
- **Computer Vision**: Image Segmentation**, Medical Imaging**, OpenCV, SegFormer, SwinIR (Super-Resolution), Canny Edge Detection, Hough Transform
- **Frameworks & Libraries**: PyTorch,TensorFlow, Scikit-learn**, LangChain, Hugging Face** Transformers
- Systems & Tools: Qdrant (Vector DB**), GLM-OCR**, Ollama (Local LLMs), Docker, **Gi**t, DSA

Work Experience

Bosch Global Software Technologies | Ai/ML Intern Bengaluru, I Jan 2026 – Present

- Built a **production-grade RAG system for AUTOSAR document intelligence** using **Python, vectorless database architecture, BM25 + dense hybrid retrieval, and Qdrant** for efficient information retrieval
- Implemented **layout-aware document parsing using GLM OCR** to extract structured **text, tables, and images**, enabling accurate type-aware understanding of complex PDFs
- Integrated **VLM and Ollama LLMs** for multimodal reasoning and response generation, ensuring **fully local, privacy-preserving**

QuadBtech Technology|Front End Developer |May 2022 – Aug2022

- Built responsive e-commerce platform with React & Bootstrap; implemented product listings, cart, and authentication features.
- Developed daily report visualizations and optimized UI performance for improved cross-device user experience.

IT Shaala | java Backend Trainee Remote | Mar 2023 – Aug 2022

- Built scalable backend apps with Spring Boot, Hibernate, REST APIs; managed MySQL and NoSQL databases in team projects.

Projects

M.Tech Major Project — Automated TB Bacilli Detection & Segmentation(Link)

In collaboration with PGI Chandigarh | Guide: Dr. Arnav Bhaskar | Clinical Expert: Dr. Nitya | Dec 2025

- Designed a two-stage deep learning pipeline for TB bacilli detection using **SwinIR** (4× **super-resolution: 100×100 → 400×400 px**) and SegFormer-B5, achieving **94.3% detection accuracy** with **4.1% false positives.**
- Built a custom dataset of **16,000 microscopy images (DDS1 + DDS3) with pixel-level annotations** across 4 WHO **severity grades,** addressing class imbalance.
- **Achieved 39.1 dB PSNR** in super-resolution (final loss: 0.00687) using progressive fine-tuning; validated on 500 clinical images from **PGI Chandigarh**.
- Developed post-processing using connected components for bacilli counting and localization**, achieving ±2.3 bacilli**/image error, improving over manual microscopy

Self-Driving Car Simulation(Link) | OpenCV, NumPy, Computer Vision (Canny, Hough Transform, ROI)

- Implemented **Canny Edge Detection** for real-time lane edge identification
- Applied **Region of Interest (ROI) masking** to focus on relevant road areas
- Used **Hough Transform** to detect lane lines from processed images
- Designed **lane smoothing and averaging algorithms** for stable detection
- Built a complete pipeline from **image input → lane detection → visualization**.

Waste Classification(Link)

- Developed a **real-time waste classification system** using **TensorFlow/Keras and OpenCV**, enabling automated categorization of waste into **Recyclable, Hazardous, Food, and Residual classes**
- Built an end-to-end pipeline integrating **webcam-based input, image preprocessing (cvzone/OpenCV), and deep learning model inference (.h5)** for real-time predictions
- Designed an interactive visualization system that provides **visual feedback by overlaying detected waste and corresponding disposal bins**, improving user guidance

Certifications & Achievements

- **GATE 2023 & 2024 Qualified** — top national engineering postgraduate entrance exam.
- **Full Stack Web Dev (MERN)** | **React.js with TypeScript & State Management** | **Data Structures & Algorithms**
- **Machine Learning & DeepLearning**